Gompertz

Parametrisation

The Gompertz distribution has log survial function

$$\log S(y) = -\frac{\mu}{\alpha}\left(\exp(\alpha y) - 1\right)$$

for response $y \geq 0$, $\mu > 0$ and $\alpha > 0$. The cummulative distribution function and the density then follows as

$$F(y) = 1 - \exp\left[-\frac{\mu}{\alpha}\left(\exp(\alpha y) - 1\right)\right]$$

and

$$f(y) = \mu \exp\left[\alpha y - \frac{\mu}{\alpha}\left(\exp(\alpha y) - 1\right)\right].$$

Link-function

The parameter μ is linked to the linear predictor η as:

$$\mu = \exp(\eta)$$

Hyperparameters

The shape parameter α is represented as

$$\alpha = \exp(S\theta)$$

and the prior is defined on θ. The constant S currently set to 0.1 to avoid numerical instabilities in the optimization, since small changes of α can make a huge difference.

Specification

- `family="gompertz"` for regression models and `family="gompertz.surv"` for survival models.
- Required arguments: y (to be given in a format by using `inla.surv()` for survival models)

Hyperparameter spesification and default values

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hyper

theta

hyperid `105101`
name `shape`
short.name `alpha`
output.name.intern `alpha_intern for Gompertz`
output.name `alpha parameter for Gompertz`
initial `-1`
fixed `FALSE`
prior `normal`
param `0 1`
to.theta `function(x, sc = 0.1) log(x) / sc`

from.theta `function(x, sc = 0.1) exp(sc * x)`

survival `FALSE`

discrete `FALSE`

link `default log neglog`

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hyper

theta1

hyperid `106101`
name `shape`
short.name `alpha`
output.name.intern `alpha_intern for Gompertz-surv`
output.name `alpha parameter for Gompertz-surv`
initial `-10`
fixed `FALSE`
prior `normal`
param `0 1`
to.theta `function(x, sc = 0.1) log(x) / sc`
from.theta `function(x, sc = 0.1) exp(sc * x)`

theta2

hyperid `106102`
name `beta1`
short.name `beta1`
output.name `beta1 for Gompertz-Cure`
output.name.intern `beta1 for Gompertz-Cure`
initial `-5`
fixed `FALSE`
prior `normal`
param `-4 100`
to.theta `function(x) x`
from.theta `function(x) x`

theta3

hyperid `106103`
name `beta2`
short.name `beta2`
output.name `beta2 for Gompertz-Cure`
output.name.intern `beta2 for Gompertz-Cure`
initial `0`
fixed `FALSE`
prior `normal`
param `0 100`

to.theta function(x) x
from.theta function(x) x

theta4

hyperid 106104
name beta3
short.name beta3
output.name beta3 for Gompertz-Cure
output.name.intern beta3 for Gompertz-Cure
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta5

hyperid 106105
name beta4
short.name beta4
output.name beta4 for Gompertz-Cure
output.name.intern beta4 for Gompertz-Cure
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta6

hyperid 106106
name beta5
short.name beta5
output.name beta5 for Gompertz-Cure
output.name.intern beta5 for Gompertz-Cure
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta7

hyperid 106107
name beta6
short.name beta6
output.name beta6 for Gompertz-Cure
output.name.intern beta6 for Gompertz-Cure

initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta8

hyperid 106108
name beta7
short.name beta7
output.name beta7 for Gompertz-Cure
output.name.intern beta7 for Gompertz-Cure
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta9

hyperid 106109
name beta8
short.name beta8
output.name beta8 for Gompertz-Cure
output.name.intern beta8 for Gompertz-Cure
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta10

hyperid 106110
name beta9
short.name beta9
output.name beta9 for Gompertz-Cure
output.name.intern beta9 for Gompertz-Cure
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

theta11

hyperid 106111

name beta10
short.name beta10
output.name beta10 for Gompertz-Cure
output.name.intern beta10 for Gompertz-Cure
initial 0
fixed FALSE
prior normal
param 0 100
to.theta function(x) x
from.theta function(x) x

survival TRUE

discrete FALSE

link default log neglog

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Example

In the following example we estimate the parameters in a simulated case

```
library(flexsurv)
library(INLA)

n <- 1000
alpha <- 1.0
intercept <- 1.1
beta <- 1.2
x <- rnorm(n, sd = 0.2)
eta <- intercept + beta*x
mu <- exp(eta)
event <- rep(1,n)
y <- rgompertz(n, rate = mu, shape = alpha)

r <- inla(y ~ 1 + x,
          family ="gompertz", data=data.frame(y, x))
r.surv <- inla(inla.surv(y, event) ~ 1 + x,
               family ="gompertzsurv", data=data.frame(y, event, x))

## should be 'small'
print(r$mlik - r.surv$mlik)
```

Notes